21.12. SOLIDIFICATION OF HIGH LEVEL LIQUID WASTES

The solidification of HLLW, followed by geologic deposition, is presently considered as the only realistic technique to create conditions for a safe long term disposal of HAW. The objectives of solidification is to immobilize the radioactive elements and to reduce the volume to be stored. The solidified product must be nondispersable (i.e. not finely divided as a powder), insoluble, and chemically inert to the storage environment, be thermally stable, have good heat conductance (this determines the maximum radioactivity and volume of the final product), be stable against radiation (up to 10^{10} Gy), and have mechanical and structural stability.

Figure 21.23 shows the options for solidification and encapsulation of HLLW. The first step is usually a calcination in which nitrates are destroyed and all metals converted to oxides. Thousands of cubic meters of HLLW have been solidified by fluidized bed calcination. However, calcine has low leach resistance, low heat conductivity, and can be dispersed in air. It is, therefore, only considered as an interim product.

Most countries have focused development work on the fixation of the active waste in borosilicate or phosphate glass. Large continuous vitrification plants producing borosilicate waste glass are located at La Hague in France and at Sellafield in the UK.

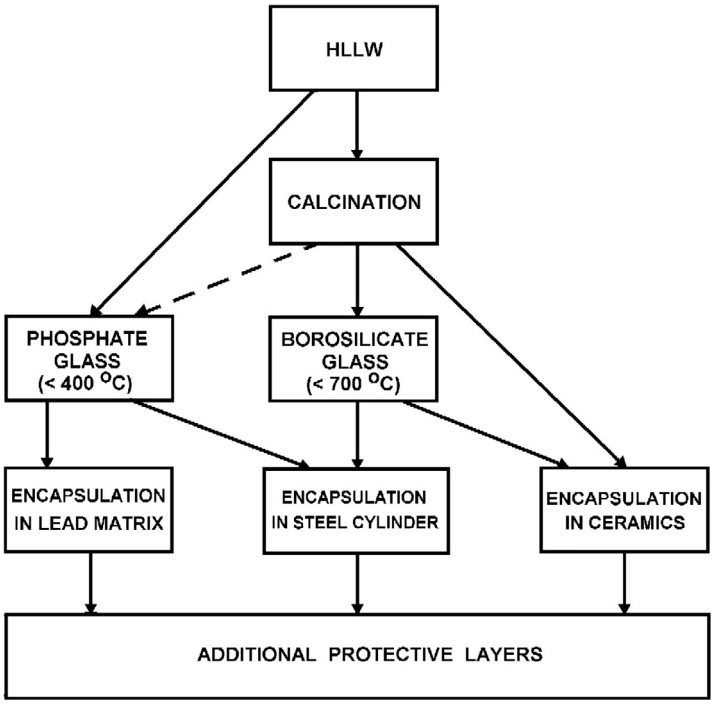


Figure 21.23 Alternative flow sheets for HLLW solidification.

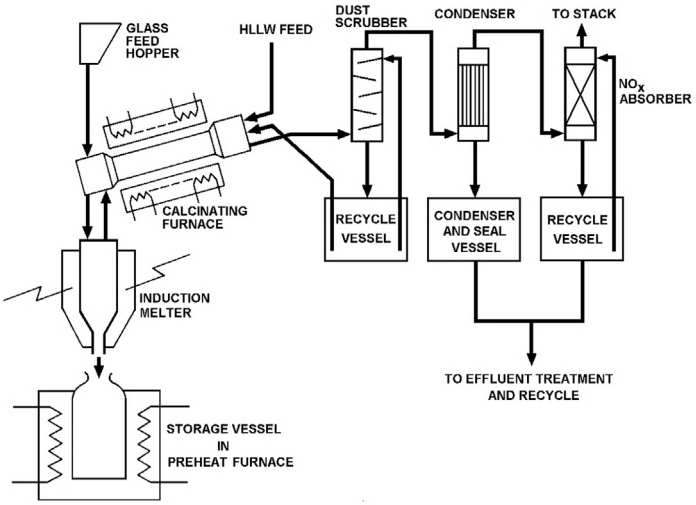


Figure 21.24 Continuous HLLW calcination and borosilicate vitrification process.

Figure 21.24 shows a rotating calciner which feeds a continuous borosilicate glass melter. Preformed and crushed borosilicate glass is continuously added near the end of calcination process. In the melter the calcines and glass mixture is heated to 1000−1200°C, leading to the formation of a homogeneous glass, which is poured into stainless steel cylinders. From a chemical standpoint (stability) the glass can contain up to 30 wt% fission product oxides, but 20% is a more normal value. Thus the waste nuclides in 1 t of LWR fuel can be contained in about 200 kg glass; the exact value depends on how long time the waste has cooled. For short cooling times a high waste content in the glass could cause it to melt and possibly crystallize (at ~700°C), which may reduce corrosion resistance because of the larger internal surface formed. The properties of some glasses are given in Table 21.11. The incorporation of the solidified waste into phosphate glass is an alternative to the use of borosilicate glass. In this case the calcination step can be bypassed. The HLLW, together with phosphoric acid, is evaporated and denitrated, and then fed to a continuous melter operating at 1000−1200°C, from where the molten glass flows into the storage pot. This process is more corrosive and produces a glass somewhat inferior to the borosilicate. Further disadvantages are that the glass recrystallizes at relatively low temperature (~400°C), and that the devitrified (crystallized) phosphate glass exhibits a rather high leach rate, 0.01−0.1 kg m^{-2} d^{-1}.

In the PAMELA process, earlier considered for use in Germany, granulated phosphate glass is incorporated into a metal alloy matrix in a steel cylinder. This offers high chemical and mechanical stability, as well as good heat conductivity (~10 W m^{-1} K^{-1} for a lead matrix; cf. 1.2 W m^{-1} K^{-1} for borosilicate glass). This decreases the central temperature and allows the incorporation of larger amounts of radioactivity (up to 35%

Table 21.11 Characteristics of some solidified high level waste products

Property	Fluidized bed calcine	Phosphate glass	Borosilicate glass
Physical form	Granular, 0.3–0.7 mm	Monolithic	Monolithic
Bulk density (kg m^{-3})	1.0–1.7	2.7–3.0	3.0–3.5
Maximum weight % FP	50	35	30
Thermal conductivity (W m^{-1} K^{-1})	0.2–0.4	1.0	1.2
Leachability (20°C, kg m^{-2} d^{-1})	1–10	10^{-5}–10^{-3}†	10^{-6}-10^{-4}‡
Maximum center temperature (°C)	550††	400*	700*
Maximum allowable heat density (kW m^{-3})	45 (#)	53	127
	38 (@)	36	92

†Devitrified (crystalline) glass has leach rates 10^{-2}–10^{-1} kg m^{-2} d^{-1}.
‡At 100°C the leach rate is ~10^{-2} kg m^{-2} d^{-1}.
††Because of risk for FP volatilization.
* Devitrifies at higher temperature.
#Forced water cooling of a cylinder; Ø 0.3 m, surface temperature ≤100°C.
@Natural air cooling of a cylinder; Ø 0.3 m.

FPs) in a single cylinder. The high heat conductivity makes it feasible to use short fuel cooling times (down to 0.5 y) and diminishes the demand for interim fuel element storage basins or HLLW storage tanks. The solidified waste glass is collected in stainless steel cylinders.

A typical waste canister may contain ~9% FPs, emit ~10 kW after 1 y, ~1 kW after 10 y, and ~0.5 kW after 40 y. They can be stored in air-cooled vaults or water-cooled pools. Before final deposition the HAW containers are reconditioned, i.e. enclosed in an additional canister of type described below. At the higher power level they are stored in water-filled pools. After >10 y air-cooled vaults may be satisfactory; forced cooling is usually considered, but with precaution for sufficient convection cooling in case of ventilation failure.

21.13. DEPOSITION IN GEOLOGIC FORMATIONS

A general consensus has developed that disposal of radioactive waste should be in the country in which the nuclear energy is produced. Thus, even if the spent fuel elements are sent from one state to another for reprocessing (like spent fuel from Japanese reactors being sent to the Sellafield reprocessing plant in the United Kingdom), the producer state (Japan) must guarantee its readiness to accept the processed waste. This condition has led to intensive national investigations of the safest way to dispose of the high level waste, either in the form of solidified HLLW, or spent fuel elements in the once-through option. In these studies geologic repositories have been the prime consideration, e.g. crystalline silicate rock, clay layers, salt domes, the sea bed, etc.